

NETWORKING - PHASE 2 (ULTRA-SHORT REVISION)

Part 1 (Chapters 1-4)

CH 1: Intro & Topologies

Network: Interconnected devices to share Data/Resources/Services.

Types: PAN (1-10m), LAN (Building), MAN (City), WAN (World).

Topologies: Bus (Backbone), Star (Hub/Switch), Ring (Circle), Mesh (Full Connect), Tree (Hierarchy).

OSI (7 Layers): 1.Phys, 2.DLL, 3.Network, 4.Trans, 5.Sess, 6.Pres, 7.App.

Mnemonic: All People Seem To Need Data Processing.

Units: App/Pres/Sess = Data; Trans = Segment; Net = Packet; DLL = Frame; Phys = Bits.

Devices: L1: Hub/Repeater; L2: Switch/Bridge; L3: Router.

TCP/IP (4 Layers): App, Transport, Internet, Network Access.

CH 2: Physical Layer

Media: Twisted Pair (Tel), Coaxial (TV), Fiber Optic (Light - Fastest).

Unguided: Radio (Broadcasting), Microwave (Satellite), Infrared (Remote).

Encoding: NRZ (No mid transition), Manchester (Mid transition).

Multiplexing: FDM (Freq), TDM (Time), WDM (Wavelength/Fiber).

Switching: Circuit (Dedicated/Tel), Packet (Internet), Message (Store & Forward).

CH 3: Data Link Layer

Flow Control: Stop-and-Wait (1 frame), Sliding Window (Multiple).

Error Detect: Parity (Count 1s), CRC (Division - Remainder 0), Checksum (Addition/Comp).

Error Control (ARQ): Stop-Wait, Go-Back-N (Resend all after loss), Selective Repeat (Resend only loss).

Access: Pure ALOHA (Anytime), Slotted ALOHA (Time slots).

IEEE: 802.3 (Ethernet), 802.4 (Token Bus), 802.5 (Token Ring).

CH 4: Network Layer

Routing: Shortest Path, Flow-Based, Distance Vector (Hop Count), Broadcast (All), Multicast (Group).

IPv4 (32-bit): A (1-126), B (128-191), C (192-223), D (224-239/Multi), E (240-255).

IPv6: 128-bit, Huge Space.

Protocols: RIP (Hop Count, max 15), OSPF (Cost), BGP (Internet).

NAT: Private to Public IP. **ICMP:** Error Messages/Ping.

Part 2 (Chapters 5-7)

CH 5: Transport Layer

TCP: Reliable, Connection-Oriented, ACK, Handshake.

3-Way Handshake: SYN -> SYN+ACK -> ACK (Connect).

4-Way Handshake: FIN -> ACK -> FIN -> ACK (Close).

UDP: Fast, Connectionless, No ACK (Video/Gaming).

Ports: HTTP: 80, HTTPS: 443, FTP: 21, SMTP: 25, DNS: 53.

Congestion: Leaky Bucket (Fixed/No burst), Token Bucket (Allows burst).

CH 6: Application Layer

DNS: Name -> IP. **FTP:** File transfer.

SMTP: Send Email. **POP3:** Download mail (clears server).

IMAP: Mail stays on server (Sync devices).

Login: Telnet (No security), SSH (Encrypted/Secure).

File Sharing: NFS (Linux), SMB (Windows).

SNMP: Network Monitoring.

CH 7: Emerging Concepts

Cloud: Store on Internet servers. **IoT:** Internet + Things.

MQTT: Broker-based, Pub-Sub, IoT sensor updates.

CoAP: Client-Server, Direct communication.

SDN: Software Defined Networking. Controller (Control Plane) manages Routers (Data Plane).

Important Comparisons

- **LAN/MAN/WAN:** Building/City/World.
- **NRZ/Manchester:** No Transition vs Mid-Transition.
- **Circuit/Packet:** Dedicated Path vs Fragments.
- **Go-Back-N/Selective:** Resend all vs Resend lost only.
- **IPv4/IPv6:** 32-bit vs 128-bit.
- **TCP/UDP:** Reliable vs Fast.
- **POP3/IMAP:** Pull vs Sync.
- **Telnet/SSH:** Insecure vs Secure.
- **MQTT/CoAP:** Broker vs Direct.

Must Remember:

OSI: App/Pres/Sess/Trans/Net/DLL/Phys

Data Units: Data -> Segment -> Packet -> Frame -> Bits

IEEE: .3(Eth), .4(Bus), .5(Ring)

RIP: 15 hops max